

People run large amounts of money with relatively unsophisticated risk management. Throughout 2008, I spoke to managers who said they had halved their risk. I would say, “Half, that’s quite a lot.” Then they would continue and say, “Yes, my leverage was four, and it is now two.” I would answer, “Do you realize volatility has gone up five times?” In terms of volatility-adjusted leverage, their risk exposure had actually gone up.

**I notice that you use VAR as a risk measurement. Aren’t you concerned that it can sometimes be very misleading regarding portfolio risk?**

*Value at Risk (VAR) can be defined as the loss threshold that will not be exceeded within a specified time interval at some high confidence level (typically, 95 percent or 99 percent). The VAR can be stated in either dollar or percentage terms. For example, a 3.2 percent daily VAR at the 99 percent confidence level would imply that the daily loss is expected to exceed 3.2 percent on only 1 out of 100 days. To convert a VAR from daily to monthly, we multiply it by the square root of 22 (the approximate number of trading days in a month). Therefore the 3.2 percent daily VAR would also imply that the monthly loss is expected to exceed 15.0 percent ( $3.2 \text{ percent} \times 4.69$ ) only once out of every 100 months. The convenient thing about VAR is that it provides a worst-case loss estimate for a portfolio of mixed investments and adapts to the specific holdings as the portfolio composition changes. There are several ways of calculating VAR, but they all depend on the volatility and correlations of the portfolio holdings during a past look-back period—and therein lies the rub. The VAR provides a worst-case loss estimate assuming future volatility and correlation levels look like the past.*

The main reason the VAR gets a bad name is because people don’t understand it. VAR does exactly what it says on the tin.

**Which is?**

It tells you how volatile your current portfolio was in the past. That is all. VAR is entirely backward looking. You have to recognize that the future will be different. If I think the world in the future will be highly volatile,